

CAMDA AMR Prediction Challenge: Predicting Antimicrobial Resistance using DNABERT-MB and DBGWAS- Derived Genomic Features

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Introduction

- Antimicrobial Resistance
- Antibiotic Susceptibility Testing
- Genomic Foundation Models
 - DNABERT, DNABERT2, DNABERT-MB
- Methods
 - Data pipeline, model development & training
- Results

Rationale

	Threat	Change in Rates or Number of Infections***			
		2020 vs. 2019	2021 vs. 2020	2022 vs. 2021	2022 vs. 2019
URGENT*	Hospital-onset CRE	▲ Increase	▲ Increase	▬ Stable	▲ Increase
	Hospital-onset Carbapenem-resistant <i>Acinetobacter</i>	▬ Stable	▬ Stable	▬ Stable	▲ Increase**
	Clinical Cases of <i>C. auris</i>	▲ Increase	▲ Increase	▲ Increase	▲ Increase
SERIOUS*	Hospital-onset MRSA	▲ Increase	▬ Stable	▼ Decrease	▬ Stable
	Hospital-onset VRE	▲ Increase	▲ Increase	▬ Stable	▲ Increase
	Hospital-onset ESBL-producing Enterobacteriales	▲ Increase	▬ Stable	▬ Stable	▲ Increase
	Hospital-onset MDR <i>Pseudomonas aeruginosa</i>	▲ Increase	▲ Increase	▬ Stable	▲ Increase

* Threat level for each pathogen, as categorized in CDC's *Antibiotic Resistance Threats in the United States, 2019*.

** There was no statistically significant difference in rate of hospital-onset carbapenem-resistant *Acinetobacter* in 2020, 2021, and 2022 when compared to the previous year. However, there was a statistically significant increase in rate of hospital-onset carbapenem-resistant *Acinetobacter* in 2022 when compared to 2019.

*** Hospital-onset rates were described using multivariable models for all threats except *C. auris*. Please note that in above table, stable indicates there was no statistically significant increase or decrease, decrease indicates a statistically significant decrease where $p < 0.05$, and increase indicates a statistically significant increase where $p < 0.05$, for all threats except for *C. auris*. Increases or decreases in *C. auris* were indicated by changes in the number of clinical cases reported nationally without hypothesis testing.

CRE - Carbapenem-resistant Enterobacteriales

(*C. auris*) - *Candida auris*

MRSA - Methicillin-resistant *Staphylococcus aureus*

VRE - Vancomycin-resistant Enterococcus

ESBL - Extended-spectrum beta-lactamase

MDR - Multidrug-resistant

- Increased antibiotic use amidst the COVID-19 pandemic reversed previous prevention efforts
- In the U.S., more than 2.8M AMR infections occur each year causing 35,000 deaths
- In 2019, there were nearly 5 million deaths associated with AMR infections

Antibiotic Susceptibility Testing (AST)

- Culture-based AST is the traditional AST method, providing a phenotype and minimum inhibitory concentration (MIC)
- Advancements in NGS sequencing technologies have provided the potential for sequencing-based AST, from:
 - **WGS of cultures (WGS-AST)**
 - metagenomic samples (metagenomic-AST)

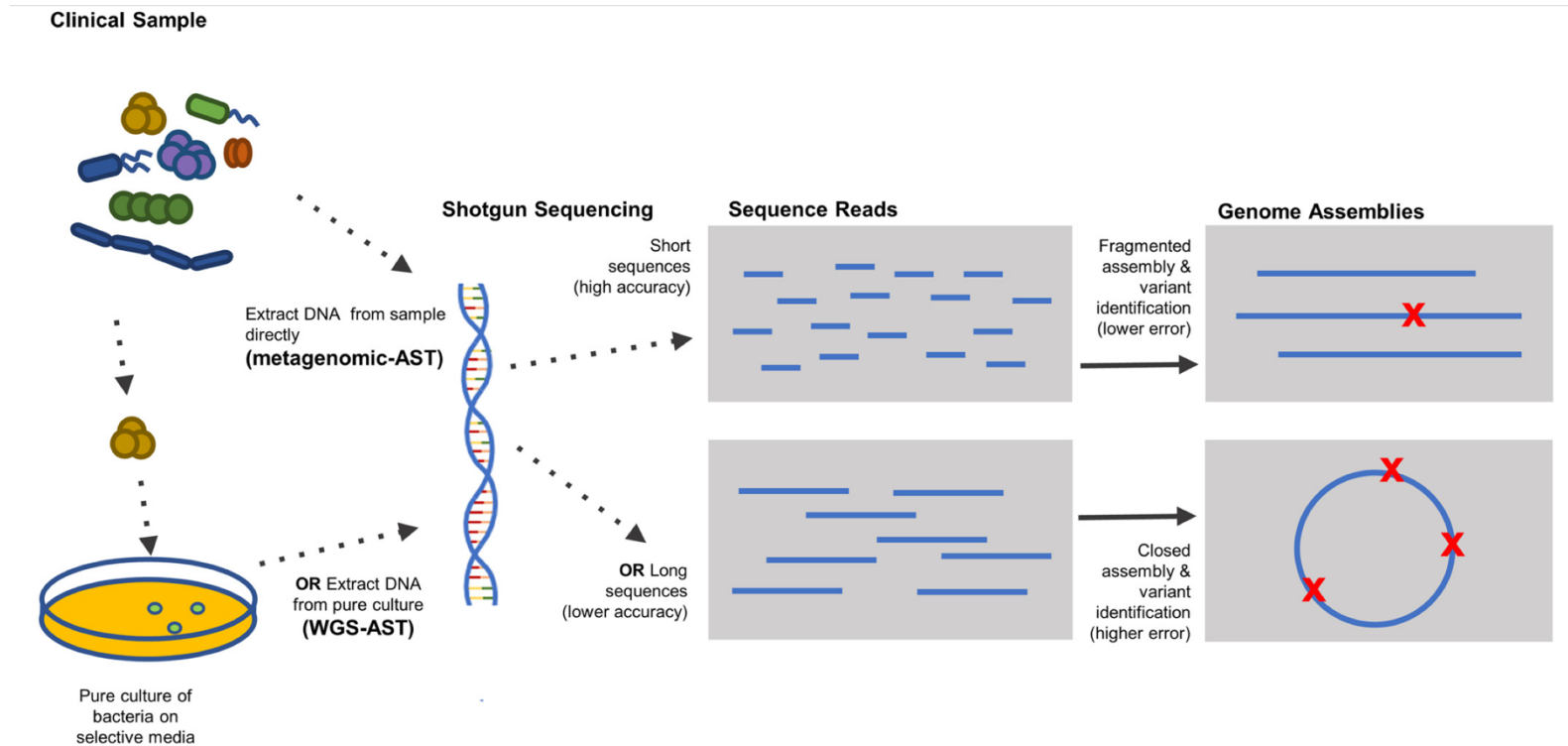


FIG 1 Overview of genome sequencing and how it is used in WGS-AST. DNA is extracted from directly from bacteria in clinical samples (metagenomics) or more commonly, from cultured bacterial colonies. Sequencing technologies fragment the DNA and then randomly sequence to produce a library of reads (stored in FASTQ files). The reads are assembled into genomic scaffolds *in silico*. Sequencing is performed either using short-read second-generation technology, which tends to produce fragmented whole-genome assemblies of high accuracy, or long-read third-generation technologies that have higher error rates but more complete assemblies. WGS-AST algorithms operate on the raw reads and/or assembled contigs.

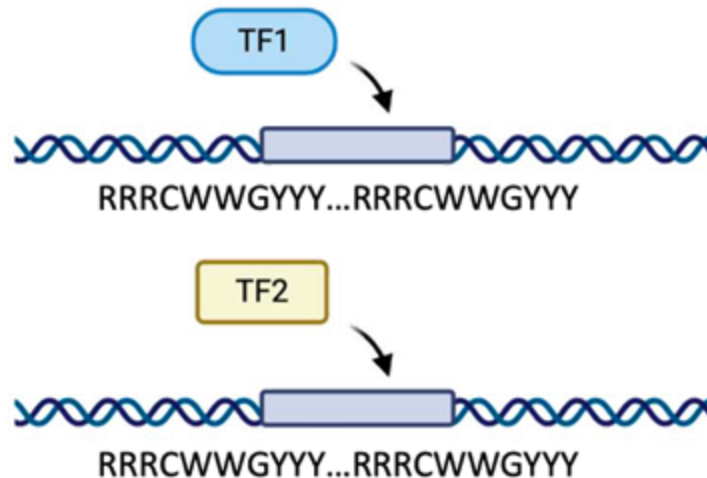
Genomic Foundation Models

- Early ML techniques had limited context
- Transformers use self-attention to capture long range dependencies
- Genomic foundation models – transformer-based models trained on vast genomic datasets

Polysemy

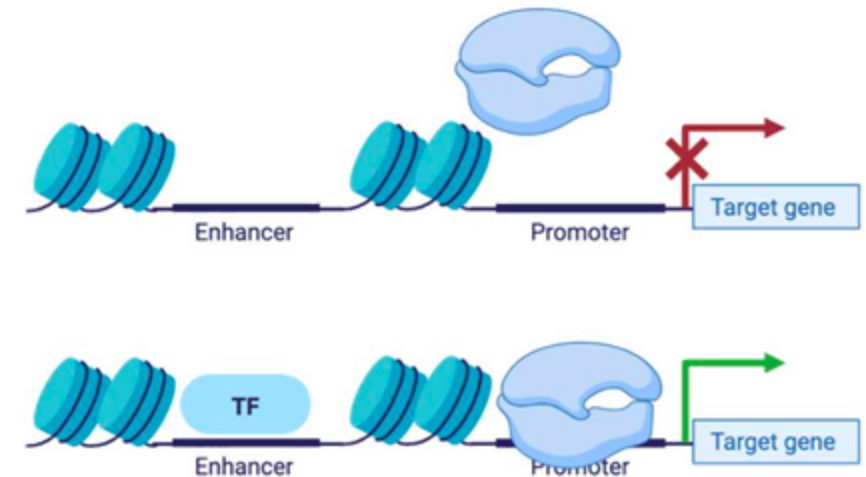
Time **flies** like an arrow.

Fruit **flies** like bananas.



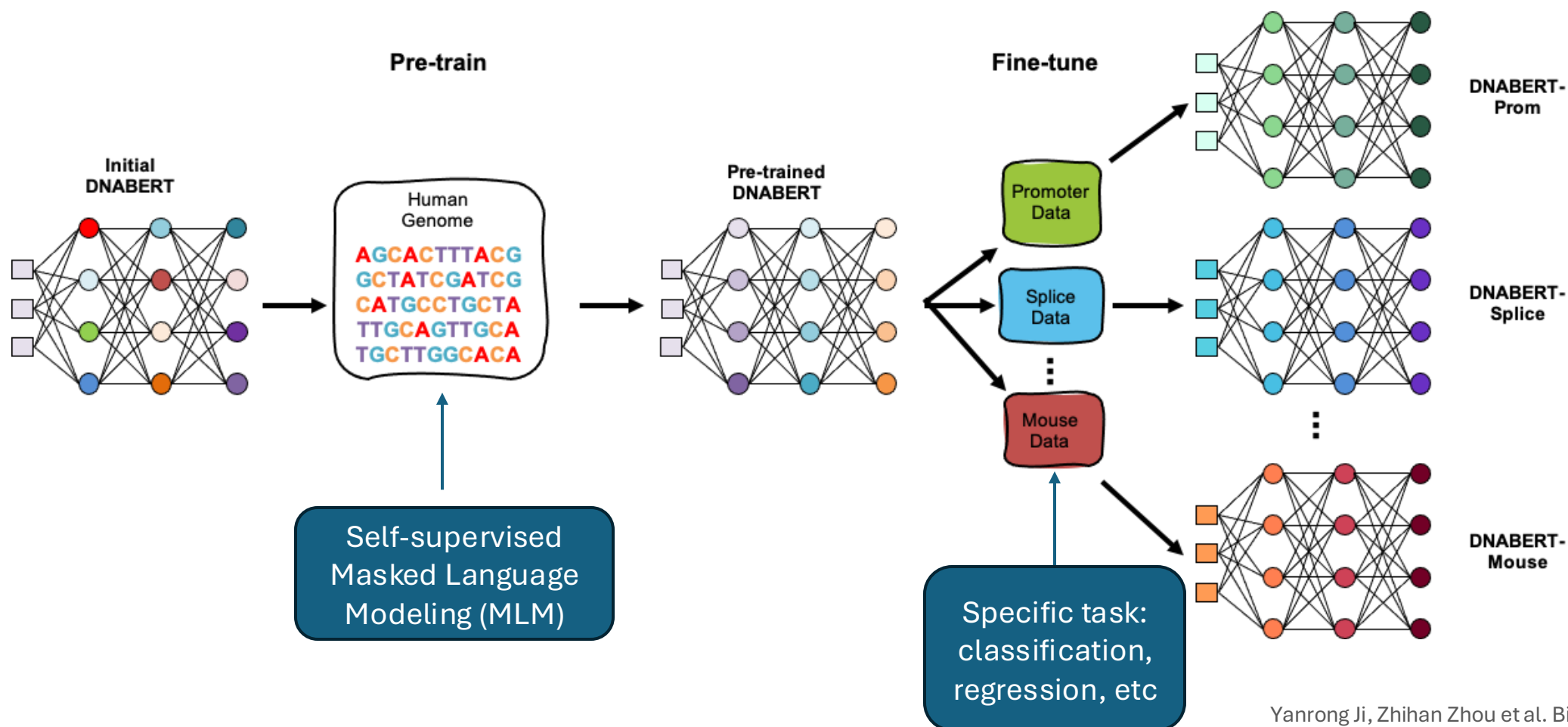
Long semantic dependency

He is **the man who** in the fullness of time and after talking to all of us who take an interest in these matters when the issue becomes important **did it**.



DNABERT

- DNABERT is a genomic foundation model pretrained on the human genome
- Based on the Bidirectional Encoder Representations from Transformers (BERT) architecture
- It adopts a pretraining-finetuning transfer learning scheme

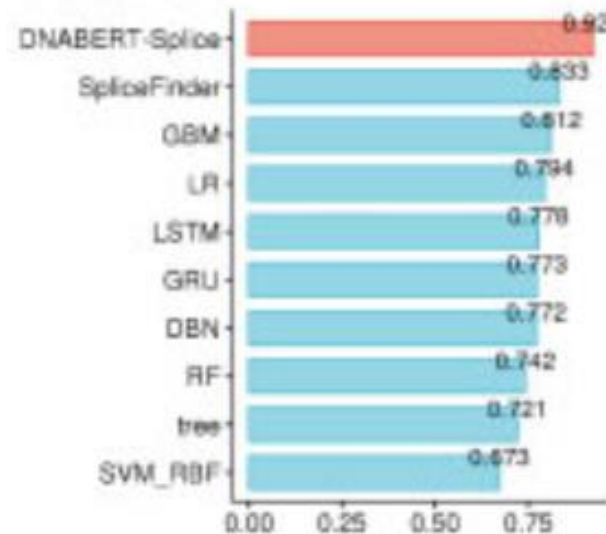


Genome analysis

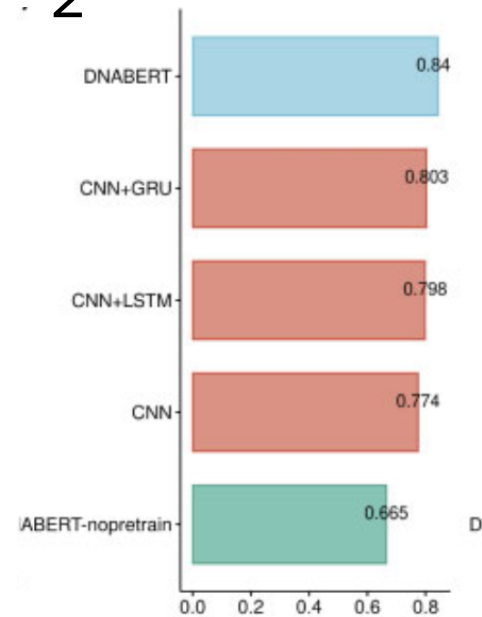
DNABERT: pre-trained Bidirectional Encoder Representations from Transformers model for DNA-language in genome

Yanrong Ji^{1,†}, Zhihan Zhou^{2,†}, Han Liu^{2,*} and Ramana V. Davuluri ^{3,*}

1



2



- DNABERT outperforms other machine learning models in splice site prediction (1) and identifying functional genetic variants (2)

DNABERT2

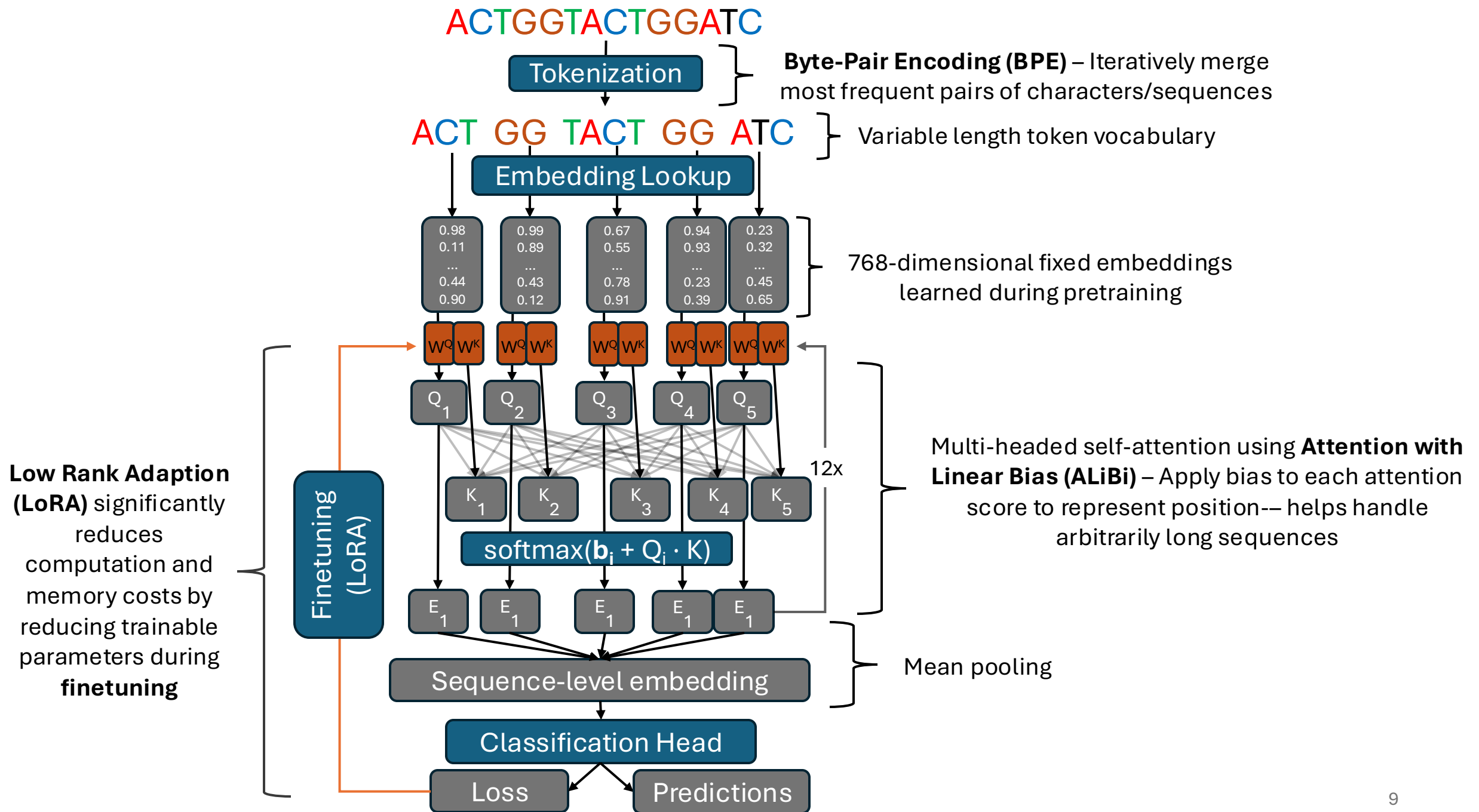
- The successor model to DNABERT
- Pretrained on genomes from six domains of life
- Incorporates Attention with Linear Bias (ALiBi) and Byte Pair Encoding tokenization
- DNABERT-MB uses this architecture, but pretrained on different sequences

$$\begin{bmatrix} q_1 \cdot k_1 & & & & \\ q_2 \cdot k_1 & q_2 \cdot k_2 & & & \\ q_3 \cdot k_1 & q_3 \cdot k_2 & q_3 \cdot k_3 & & \\ q_4 \cdot k_1 & q_4 \cdot k_2 & q_4 \cdot k_3 & q_4 \cdot k_4 & \\ q_5 \cdot k_1 & q_5 \cdot k_2 & q_5 \cdot k_3 & q_5 \cdot k_4 & q_5 \cdot k_5 \end{bmatrix} + \begin{bmatrix} 0 & & & & \\ -1 & 0 & & & \\ -2 & -1 & 0 & & \\ -3 & -2 & -1 & 0 & \\ -4 & -3 & -2 & -1 & 0 \end{bmatrix} \cdot m$$

Attention with Linear Bias (ALiBi) calculation – the further in the past a given key is, the higher a value is subtracted from the attention value– this replaces positional embeddings

Iteration	Corpus	Vocabulary
0	AACGGCACTATATA	{A,T,C,G}
1	AACGGCACTATATA	{A,T,C,G,TA}
2	AACGGCACTATA	{A,T,C,G,TA,AC}
3	AACGGCACTATA	

Byte Pair Encoding (BPE) – common sequences of characters are iteratively merged to create a vocabulary of variable-length tokens



Workflow

1. DNABERT-MB Pretraining
 1. Data
 2. Methodology
2. Data Pipeline
 1. Data (train/test/dev and independent validation)
 2. Assembly
 3. DBGWAS
 4. Finetuning dataset creation
 5. Model sets
3. Finetuning/Development of classifier
 1. Finetuning
 2. Classification head
 3. Majority voting

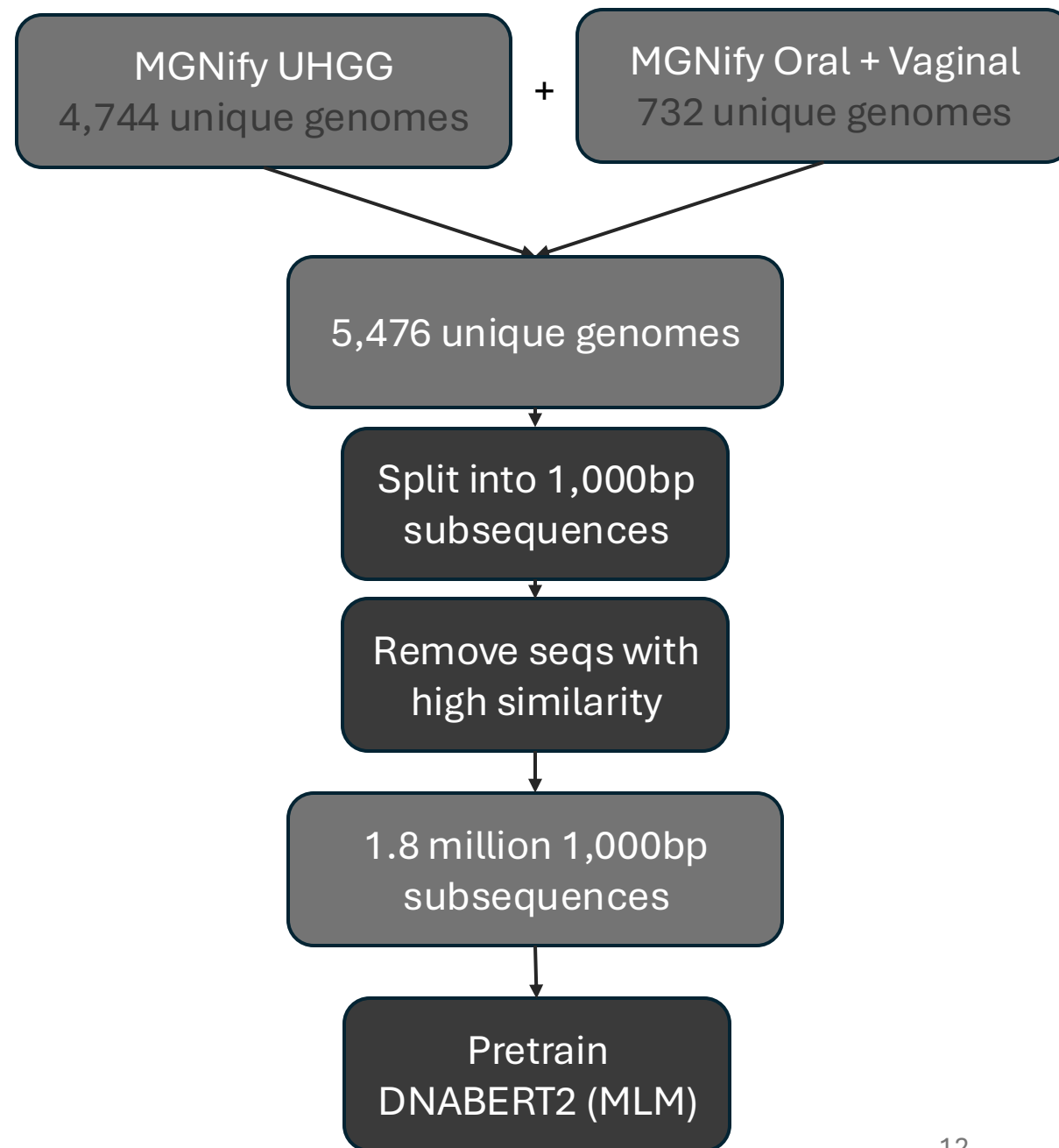
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DNABERT-MB

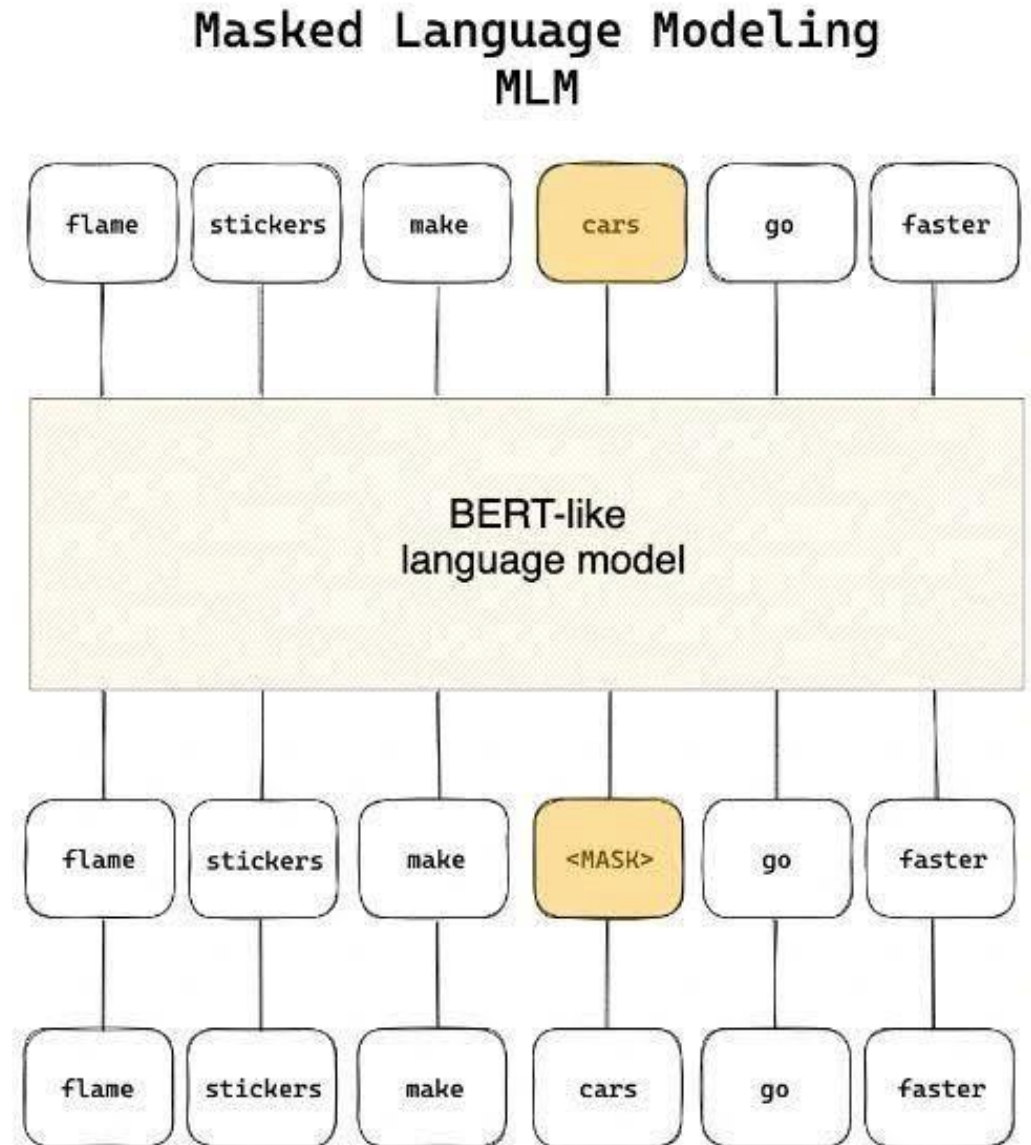
Pretraining Data

- DNABERT-MB was pretrained on 1,000bp sequences **only from the human microbiome**
- Including metagenome assembled genomes (MAGs) and cultured isolates



Pretraining Methodology

- Masked Language Modeling (MLM)
- Self-supervised learning
- Tokens are randomly masked
- Model must predict masked tokens

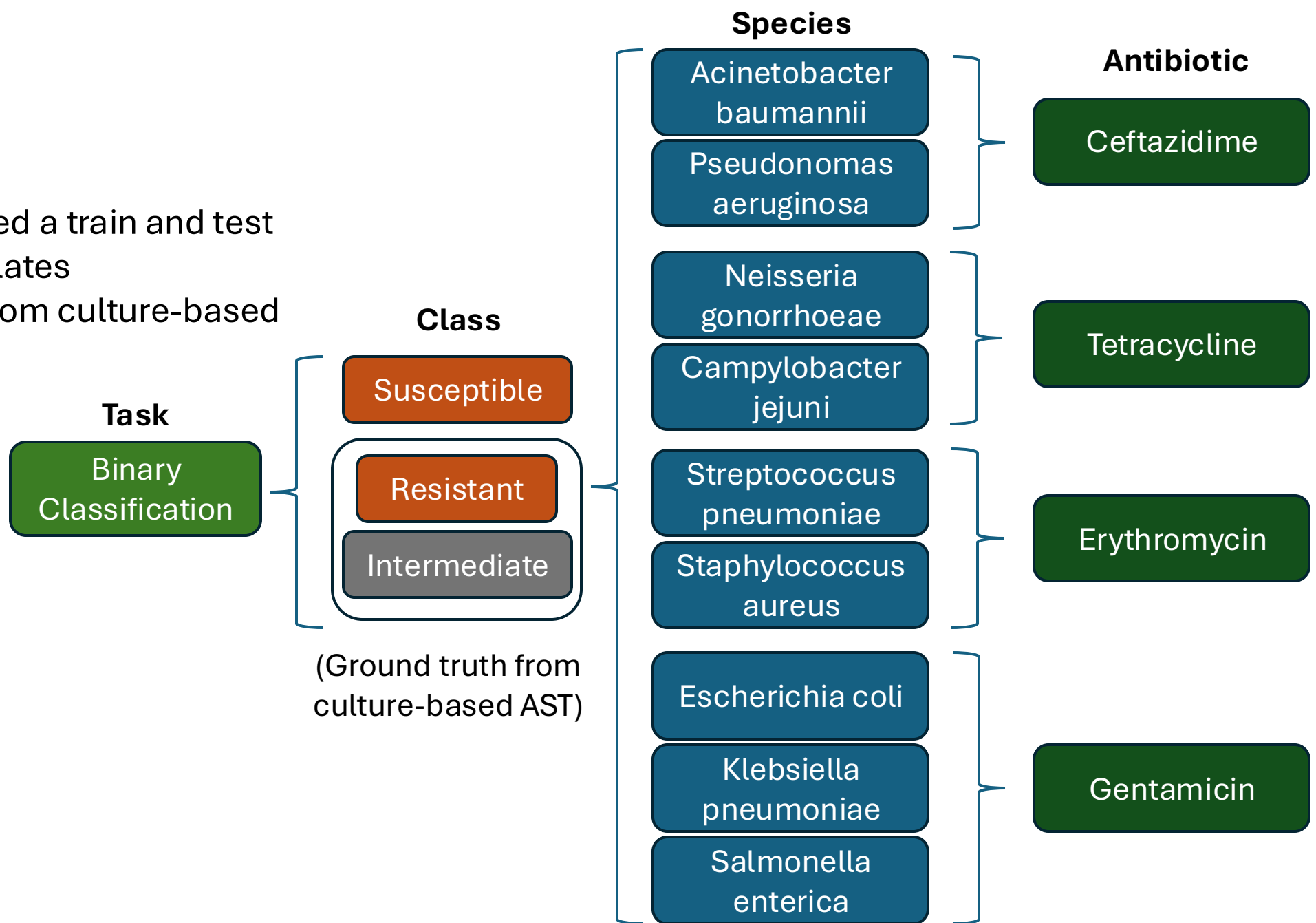


Workflow

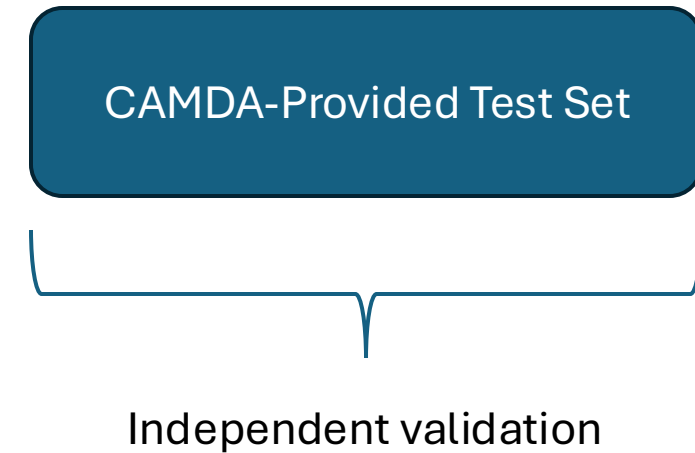
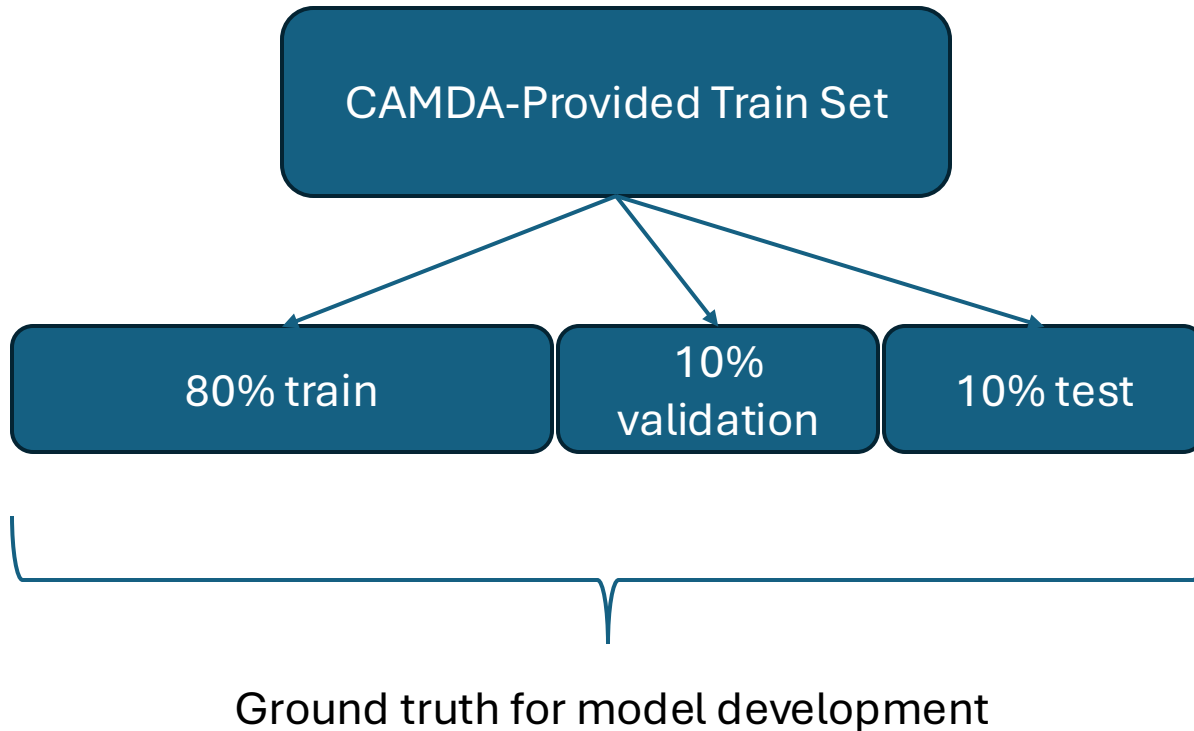
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Data

- CAMDA provided a train and test set of WGS isolates
- Ground truth from culture-based AST



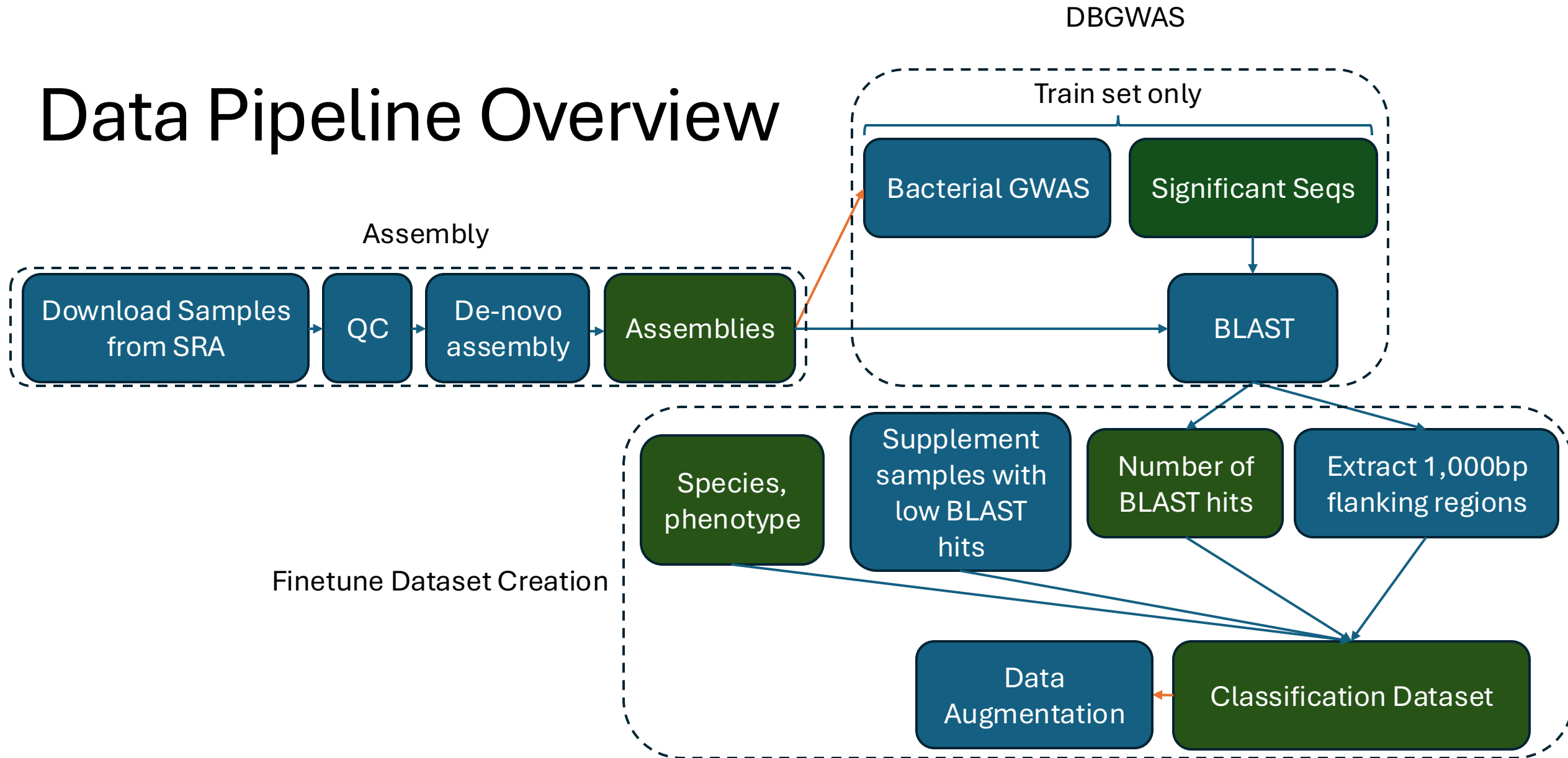
Model development split and independent validation



Workflow

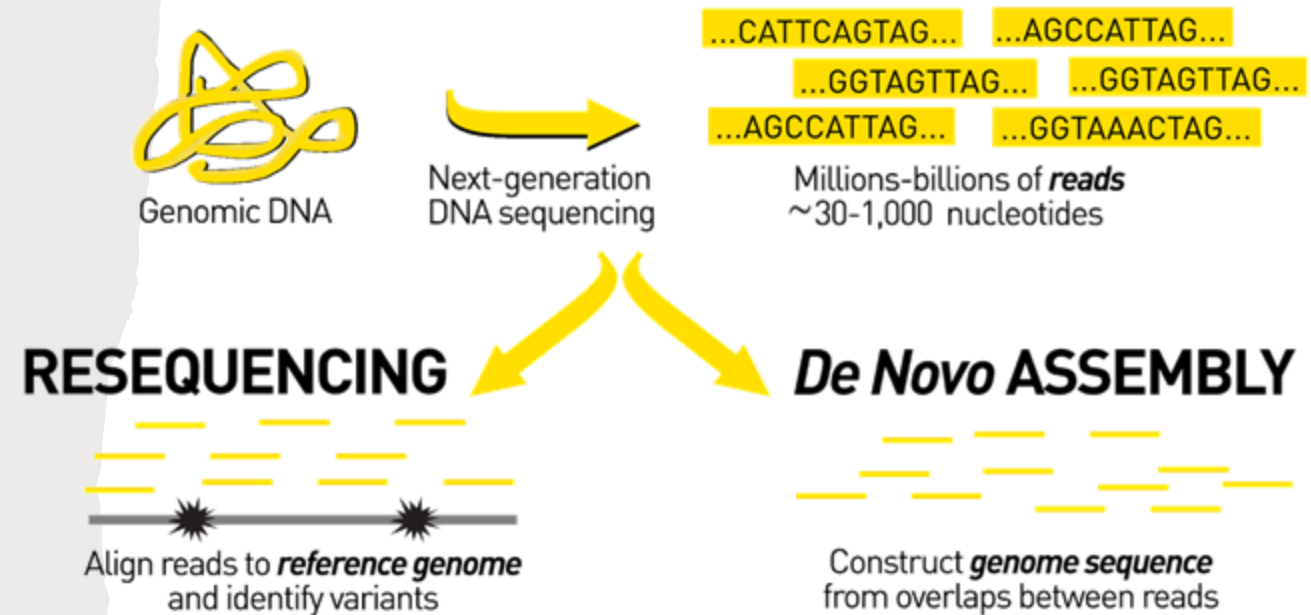
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Data Pipeline Overview



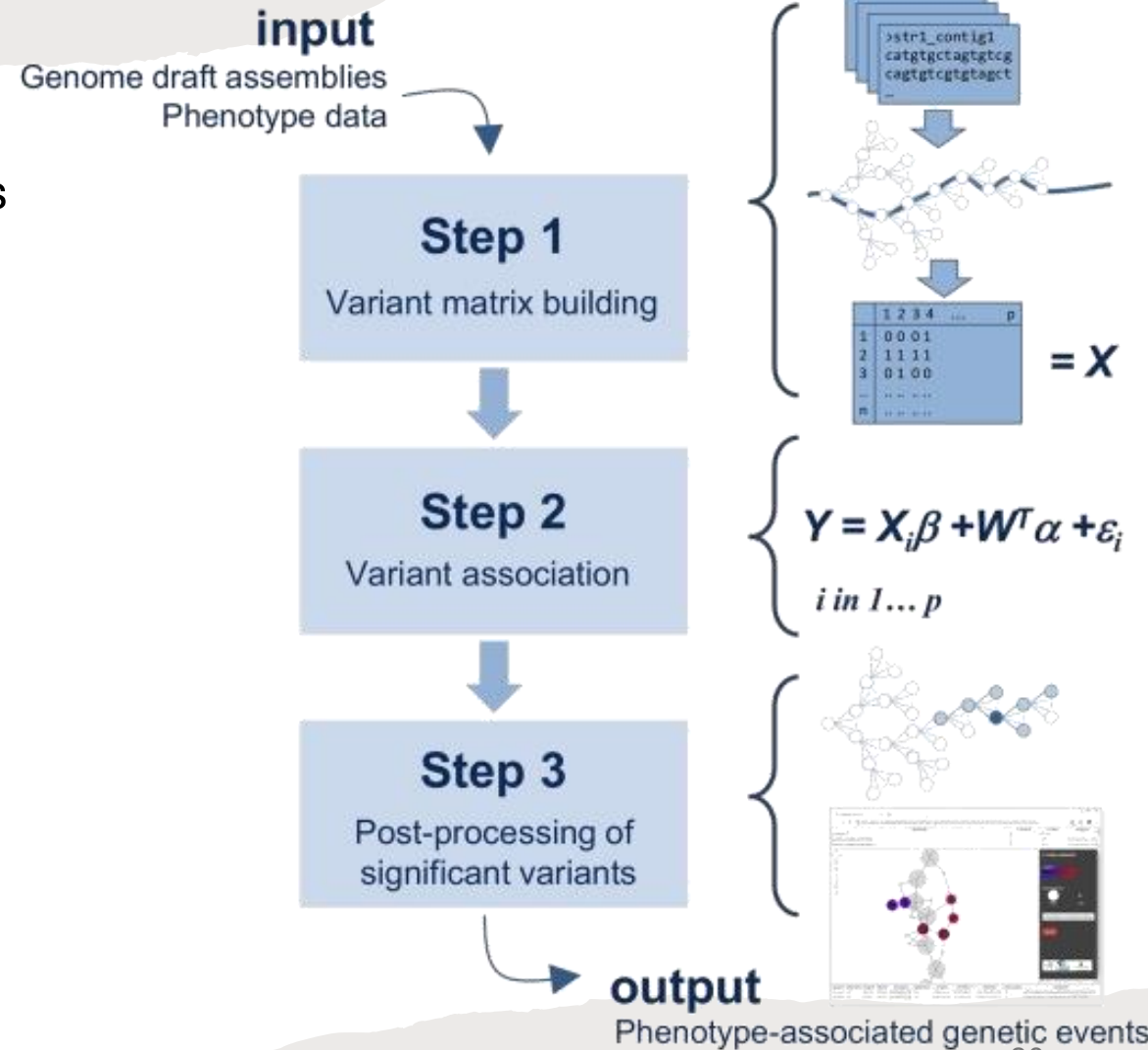
Assembly

- Bacteria acquire AMR through horizontal gene transfer and spontaneous mutations
- Many AMR genes are located on mobile genetic elements
- Finding an all-inclusive reference genome is not possible
- De novo assembly with SPAdes was chosen

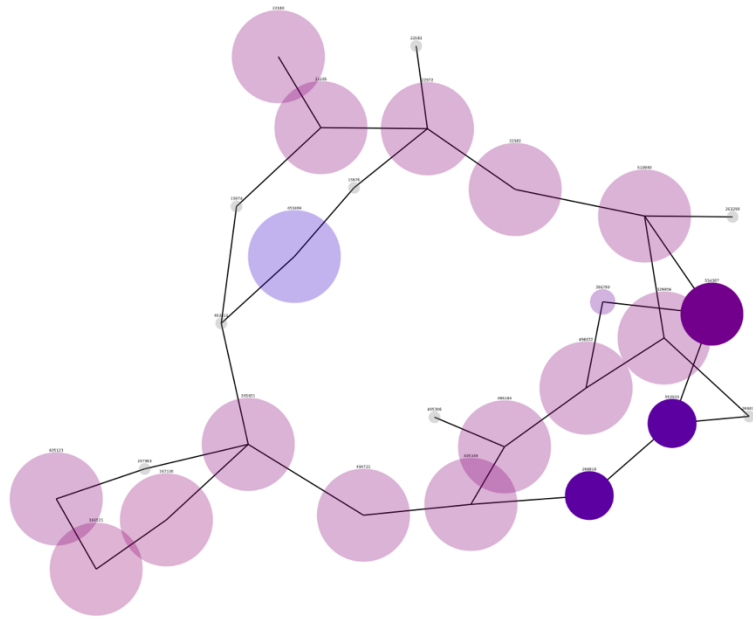


De-Brujin Genome Wide Association Study (DBGWAS)

- Traditional GWAS needs reference genomes and focuses on SNPs/indels
- Bacterial GWAS tools:
 - Alignment-free
 - Capture broader variation like MGEs and gene presence/absence
- DBGWAS:
 - k-mer based approach
 - Builds cDBGs to extract shared genetic elements
 - These are associated with phenotype using linear mixed models
 - Validated by finding AMR determinates

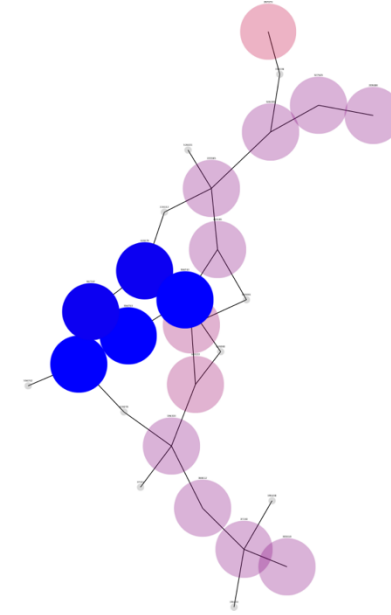


A



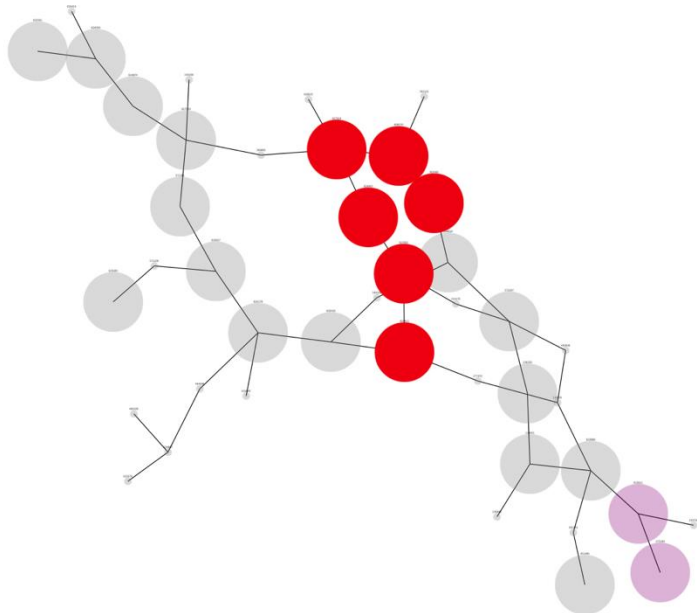
- Plasmid

B



- Resistance-associated polymorphism

C



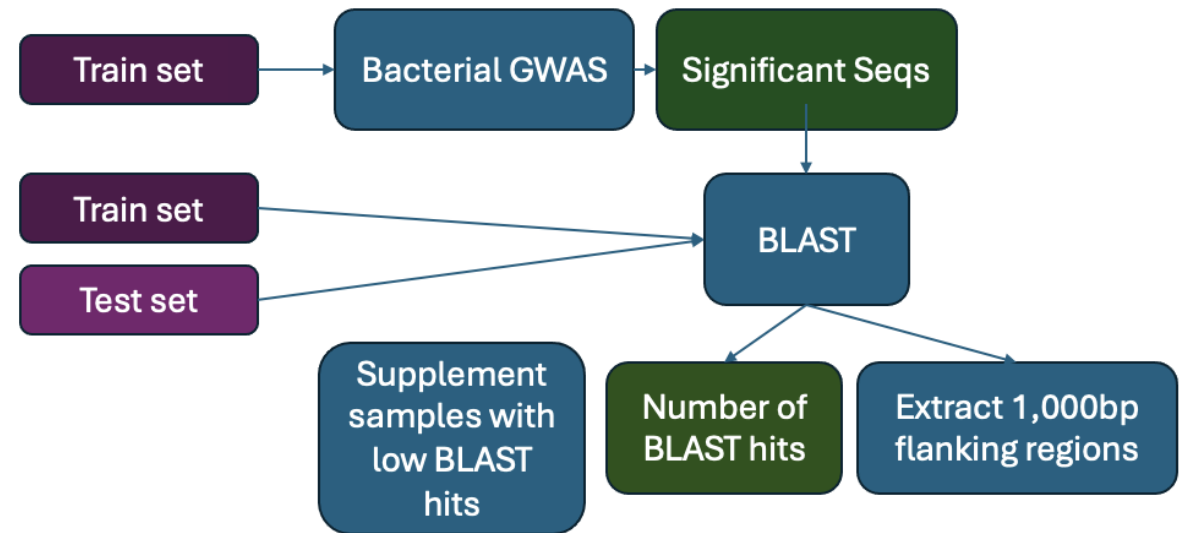
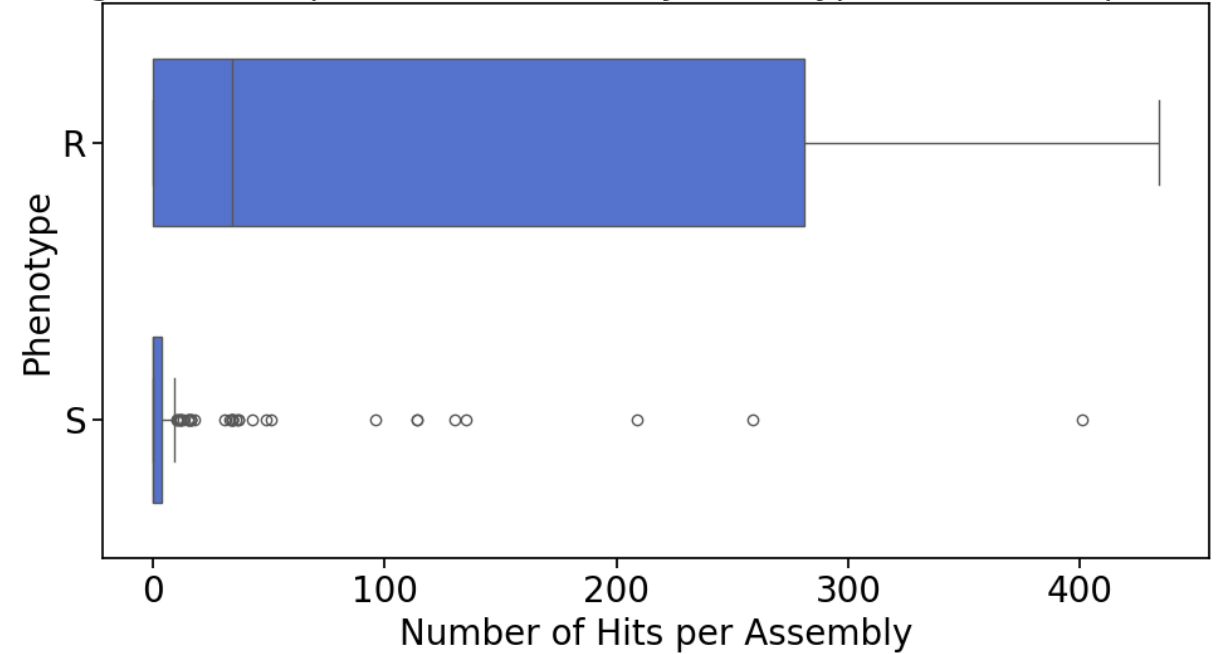
- Susceptible-associated polymorphism

Subgraphs generated from DBGWAS where blue nodes represent resistant-associated genetic events and red represent susceptible. Transparency represents statistical significance. Size represents allele frequency. Bubbles in the graph are where shared continuous sequences split off into two paths

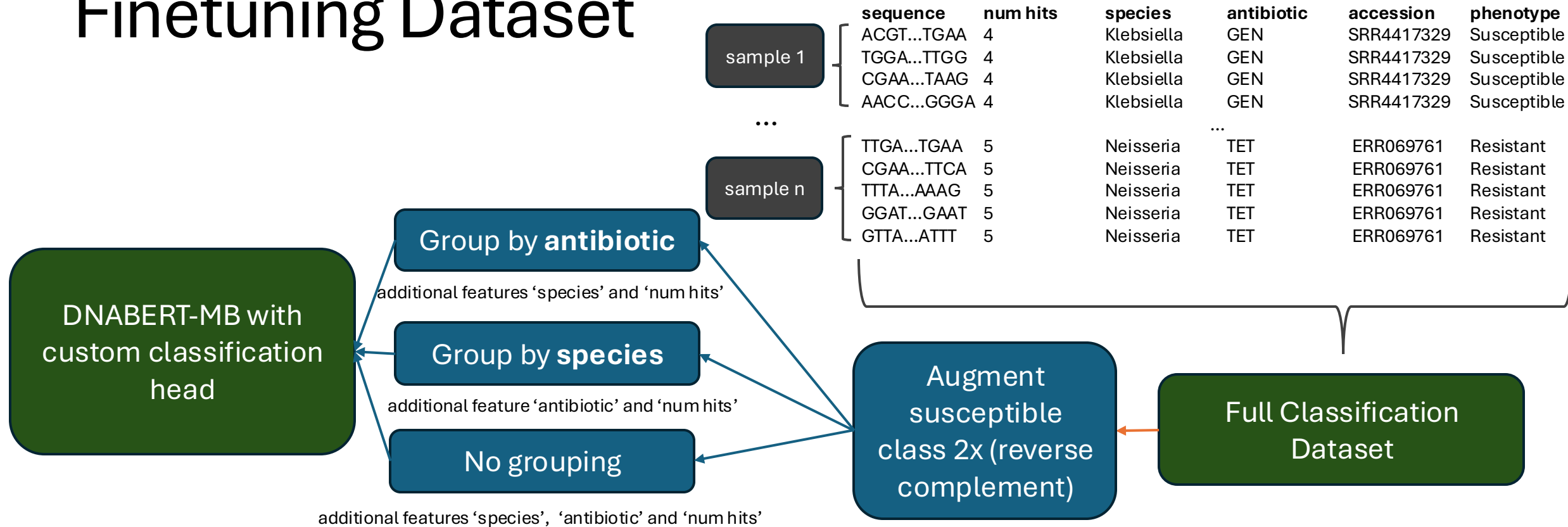
BLAST

- Genetic sequences were filtered by p-value and BLASTed back to original assemblies
- Number of BLAST hits taken as an additional feature
- 1,000bp flanking regions were taken around each hit

Significant Sequence BLAST Hits by Phenotype at DBGWAS $p < 5e-12$



Finetuning Dataset



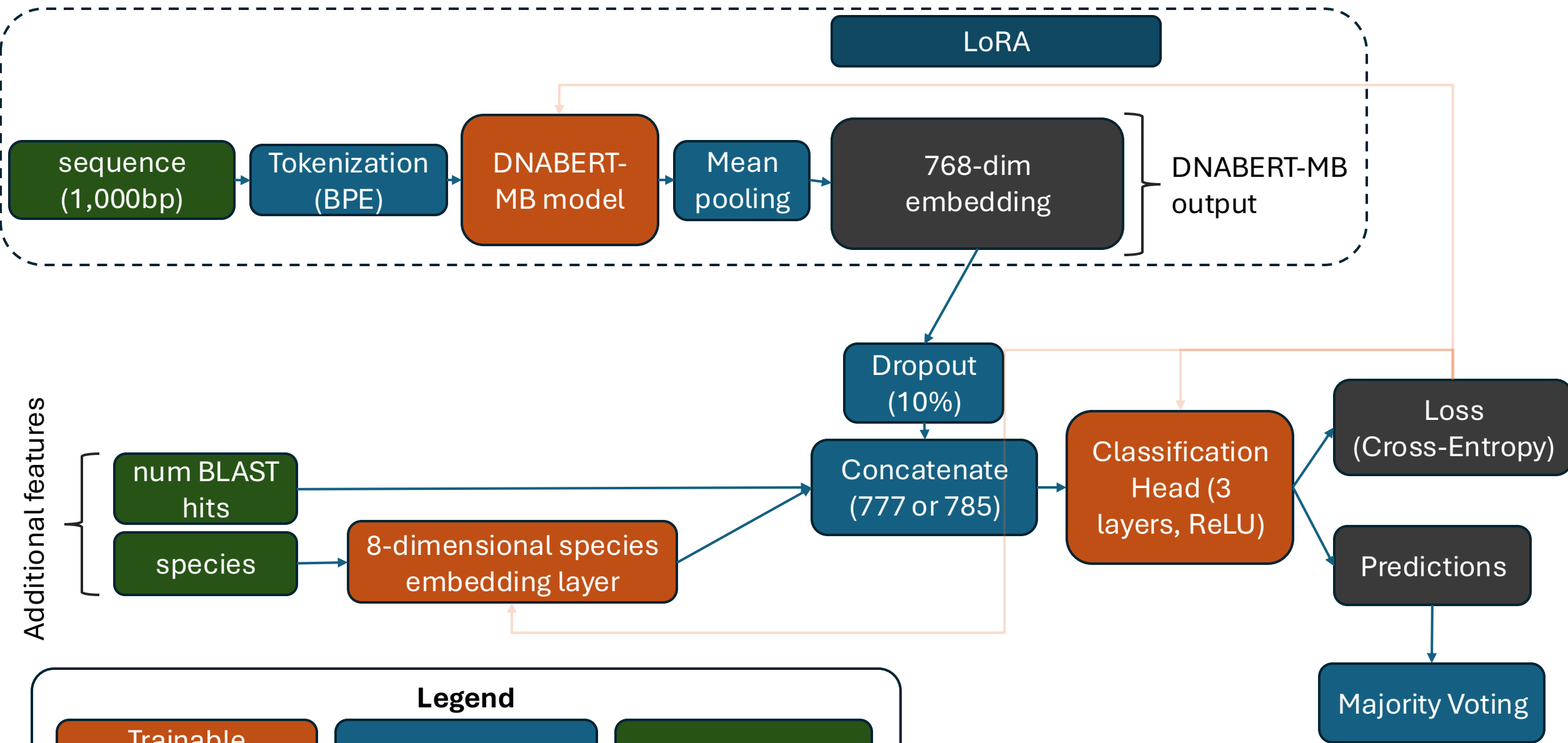
- We now have a list of **sequences** each belonging to a bacterial **sample** of a particular **species** treated on a certain **antibiotic** resulting in a binary **phenotype**

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Finetuning (4x)

DNABERT-MB Forward Pass



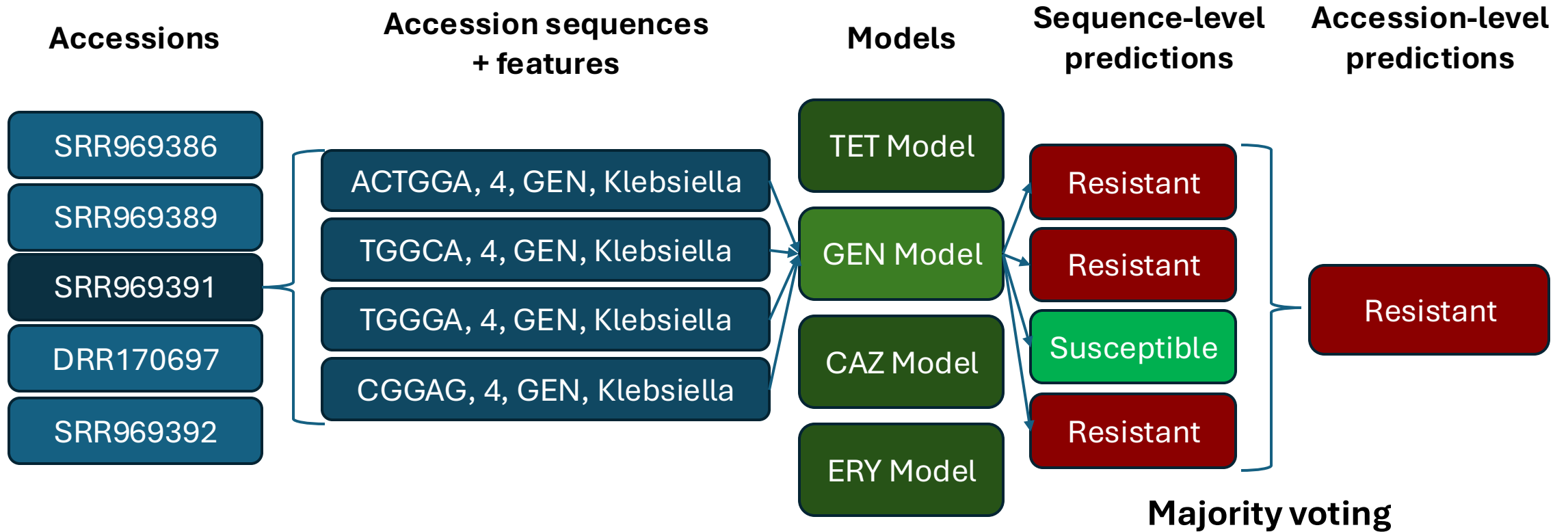
Legend

Trainable
Network

Process

Feature

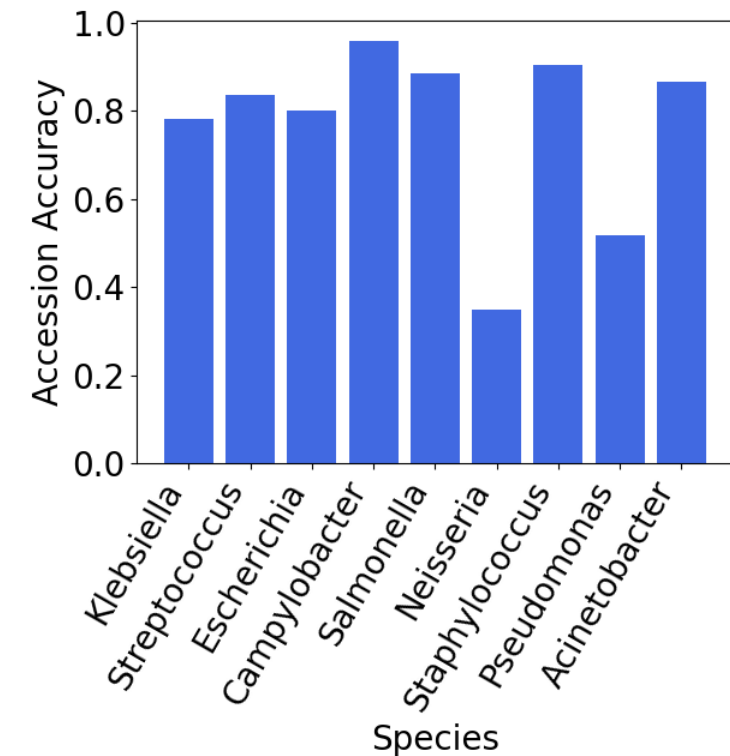
Majority voting (consensus inference)



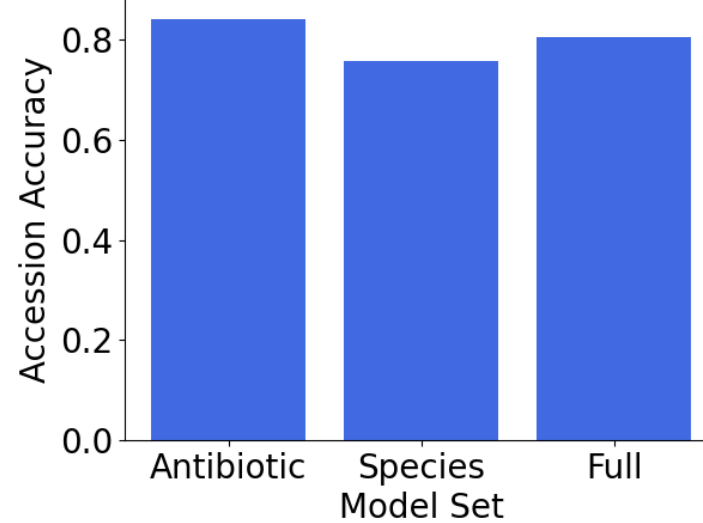
- Inference dataset grouped by antibiotic and accession number
- Final prediction made by taking most frequent prediction per accession
 - ‘accession-level’ predictions

Results

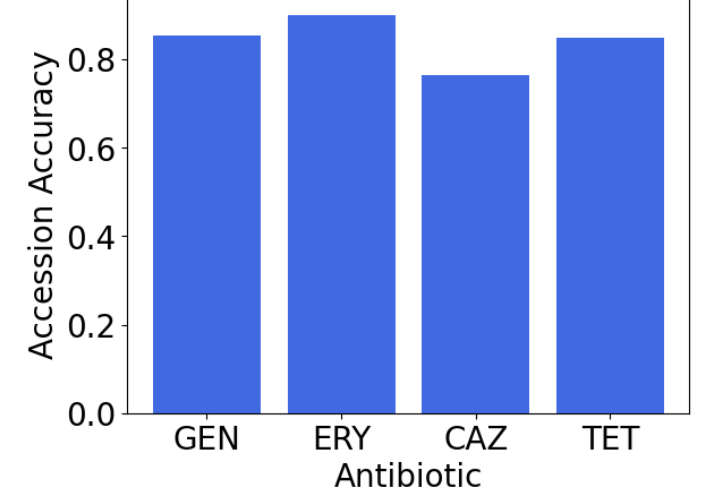
Per-Species Models Accession Accuracy



Overall Accession Accuracy by Model Set

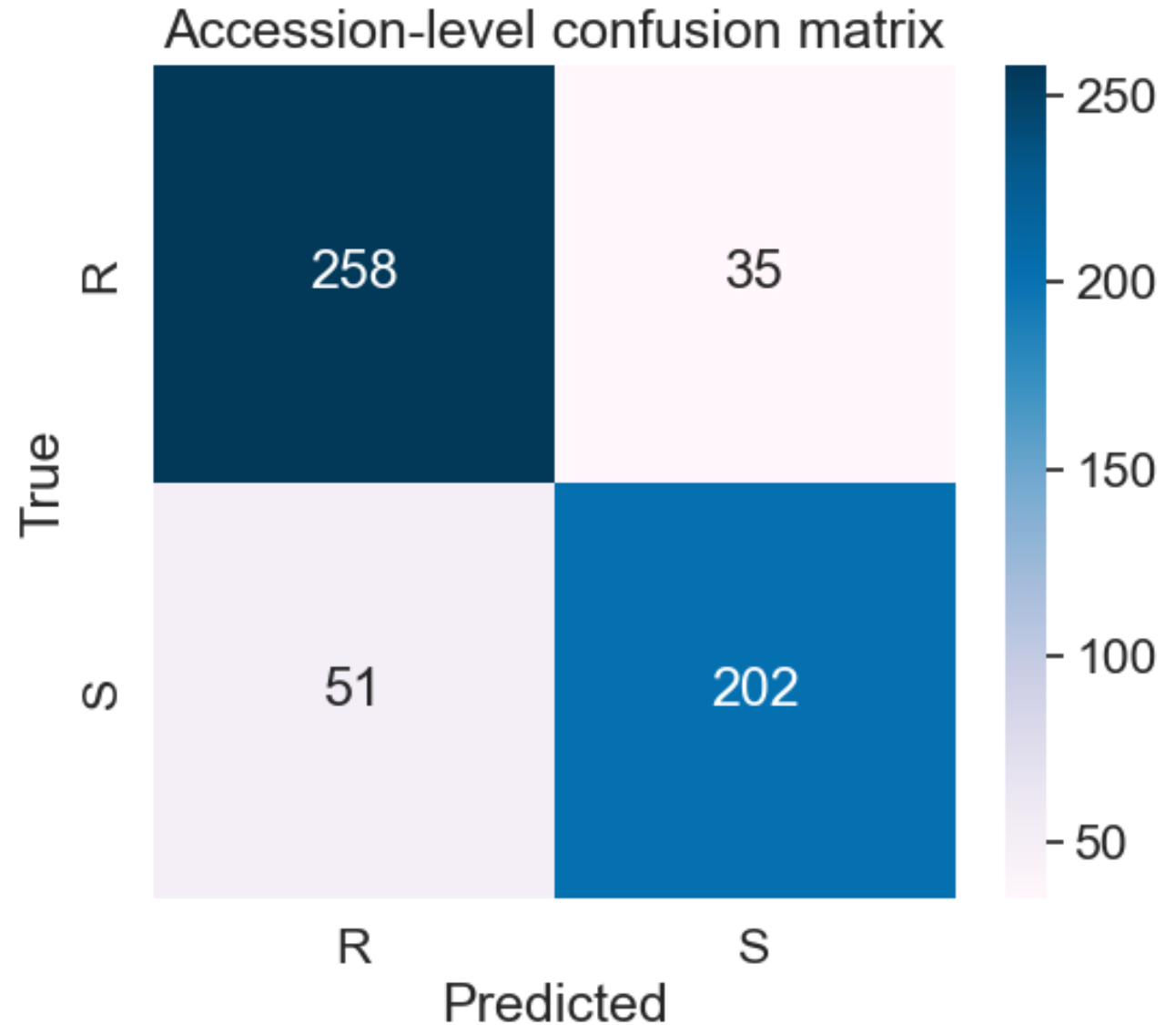


Per-Antibiotic Models Accession Accuracy



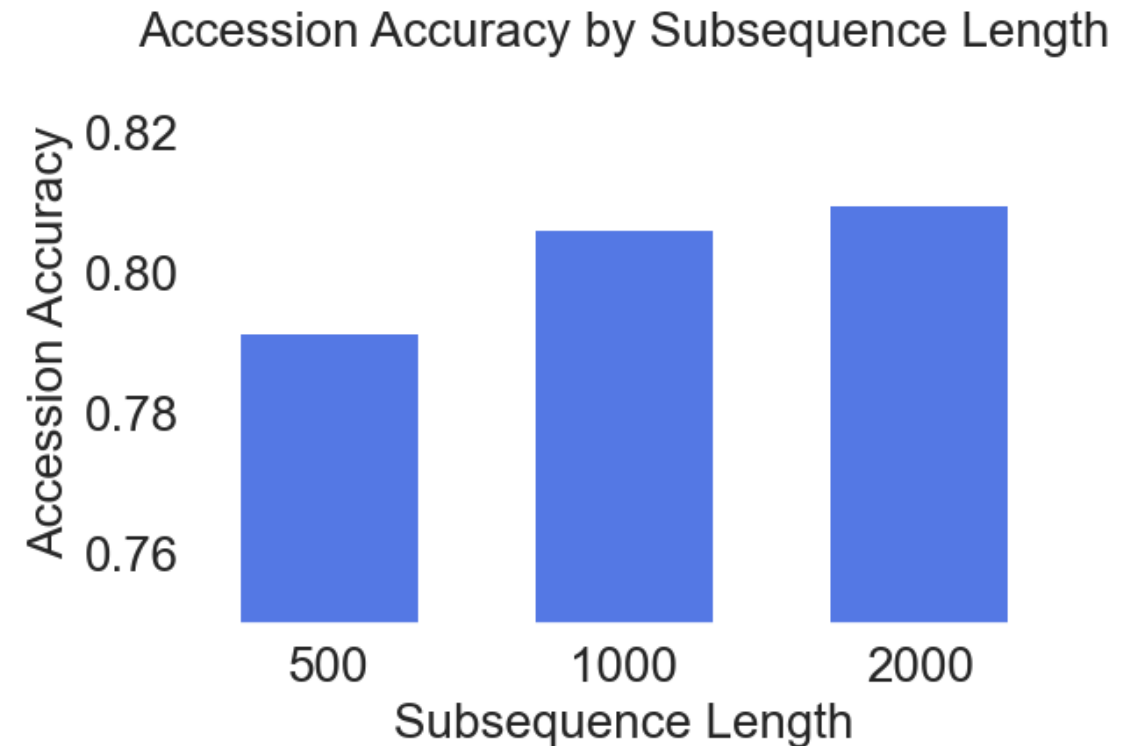
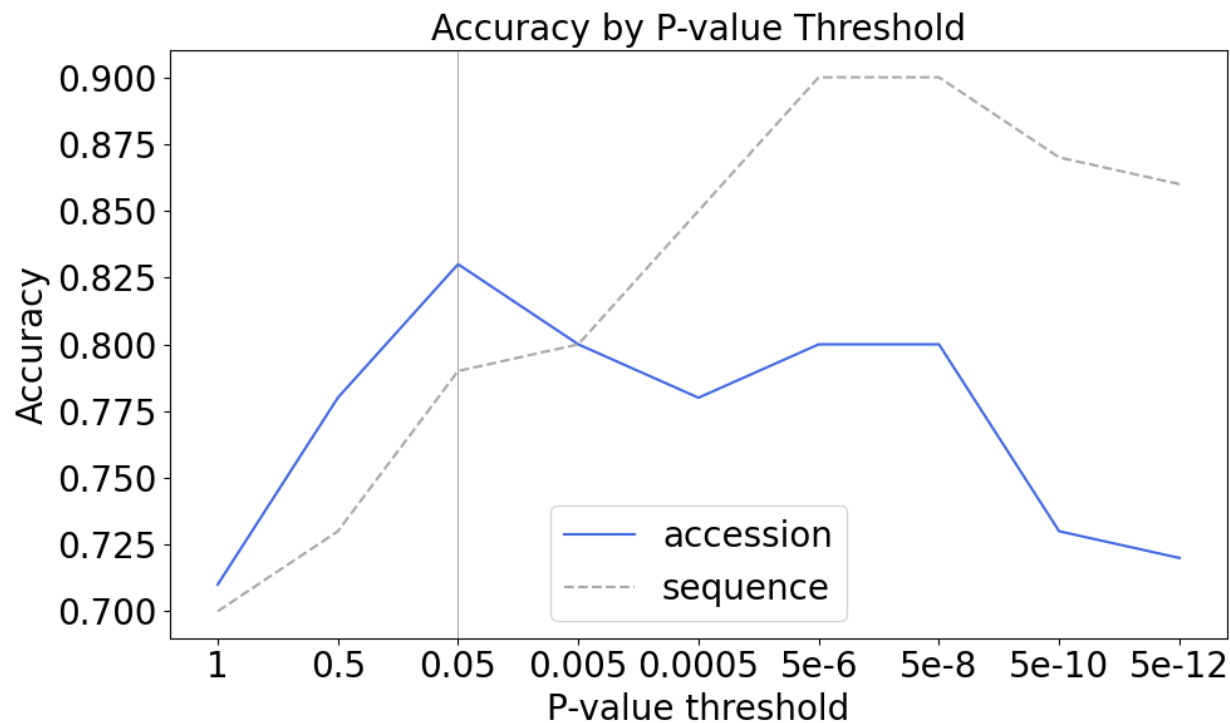
Per-Antibiotic Model Performance

- CAMDA test set accuracy: **82.8%**, f1-score 0.829
- Original DNABERT2 pretrained model: 3% accuracy decrease



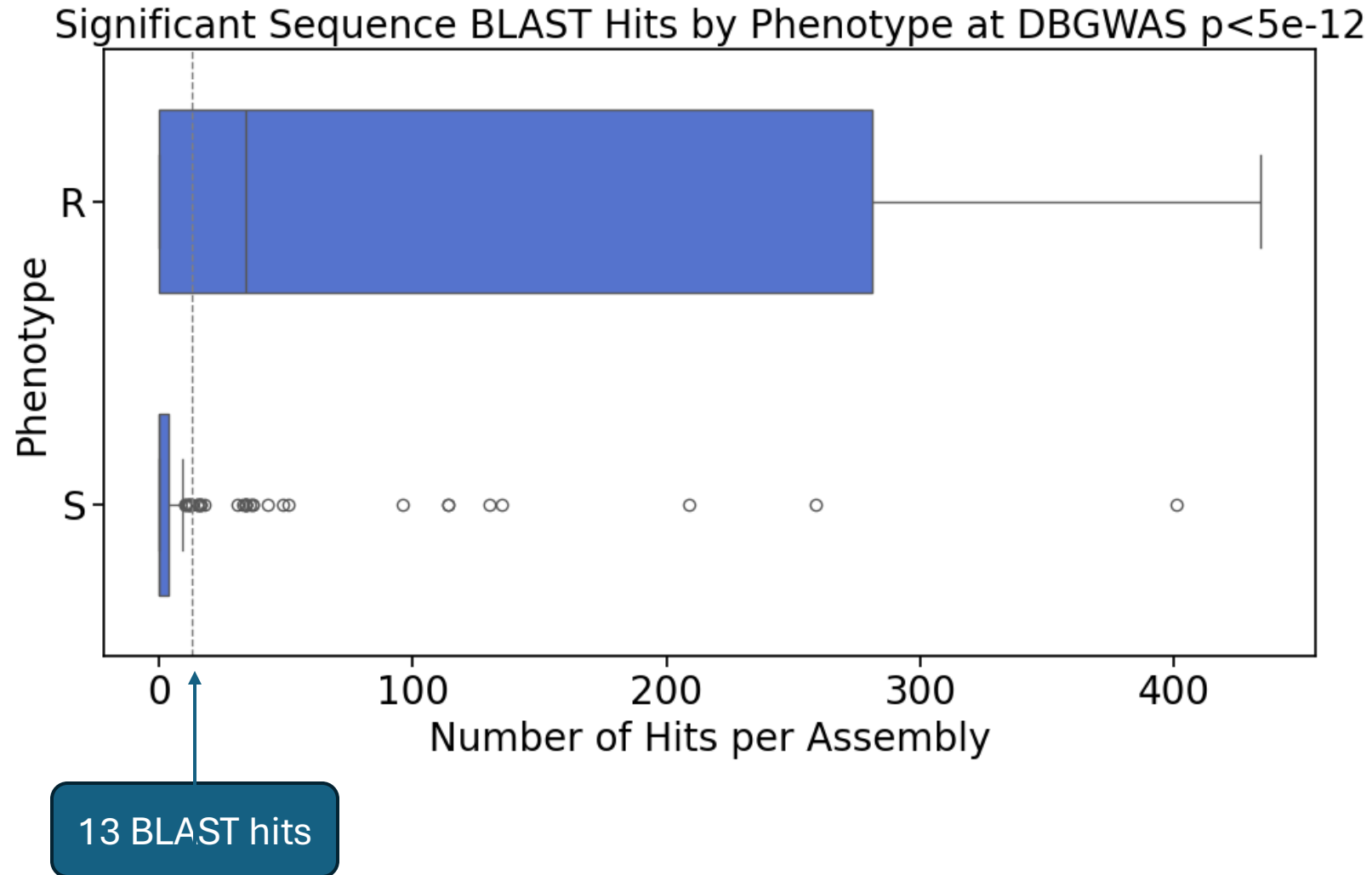
Data Pipeline Parameters

- Accession-level accuracy peaks at DBGWAS $p < 0.05$
- 2000bp length has minimal performance improvement



DNABERT- MB Abalation

A threshold of 13 BLAST hits gave the best performance with an accuracy of **71.6%** using DBGWAS $p < 5e-12$



Discussion

- Number of BLAST hits alone is discriminative feature, but DNABERT-MB is required for better performance
- Performance increase of DNABERT-MB from DNABERT2 highlights importance of domain-specific pretraining
- Finetuning enables model to focus on resistance **mechanism** rather than species-specific features
 - Reinforces idea that resistance genes are not always species-specific (Barcan et al., 2025)

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- ◆ NIH 1R01DA062257 (PI: Tang)



Stony Brook Cancer Center

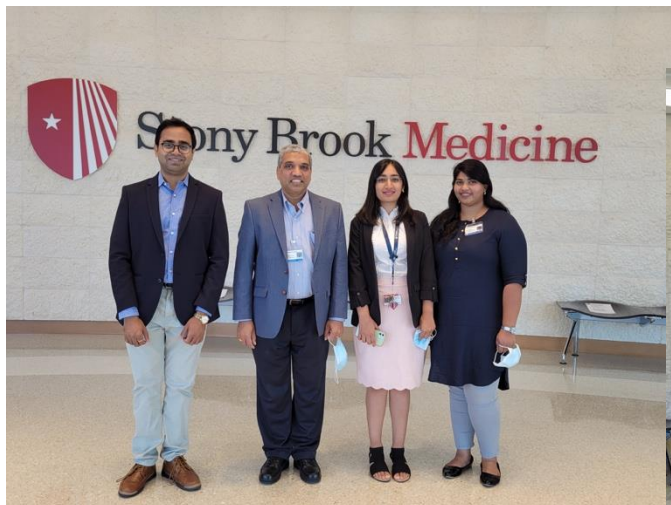
Han Liu, PhD
Northwestern CS



Zhihan Zhu, PhD



Yanrong Ji, PhD





Questions?

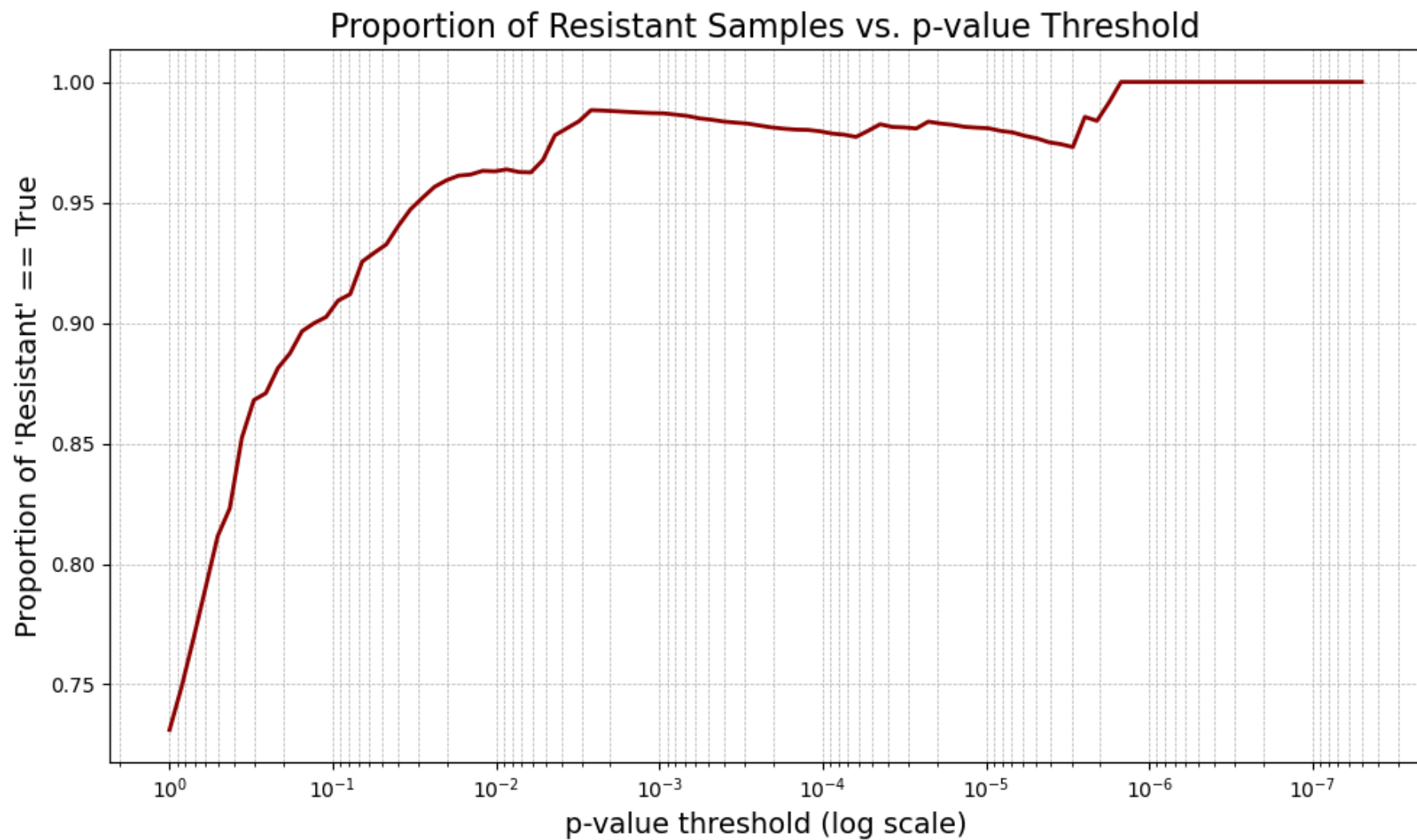


Jack Vaska

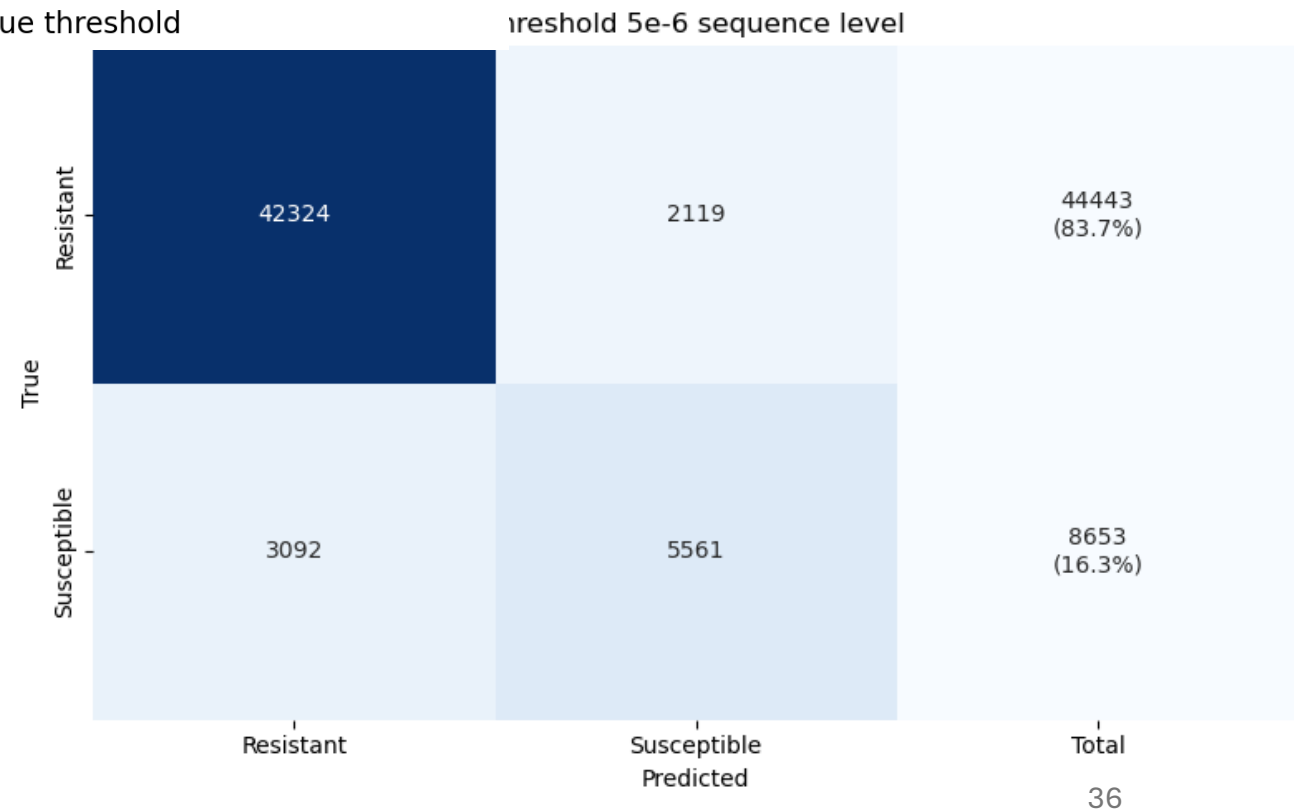
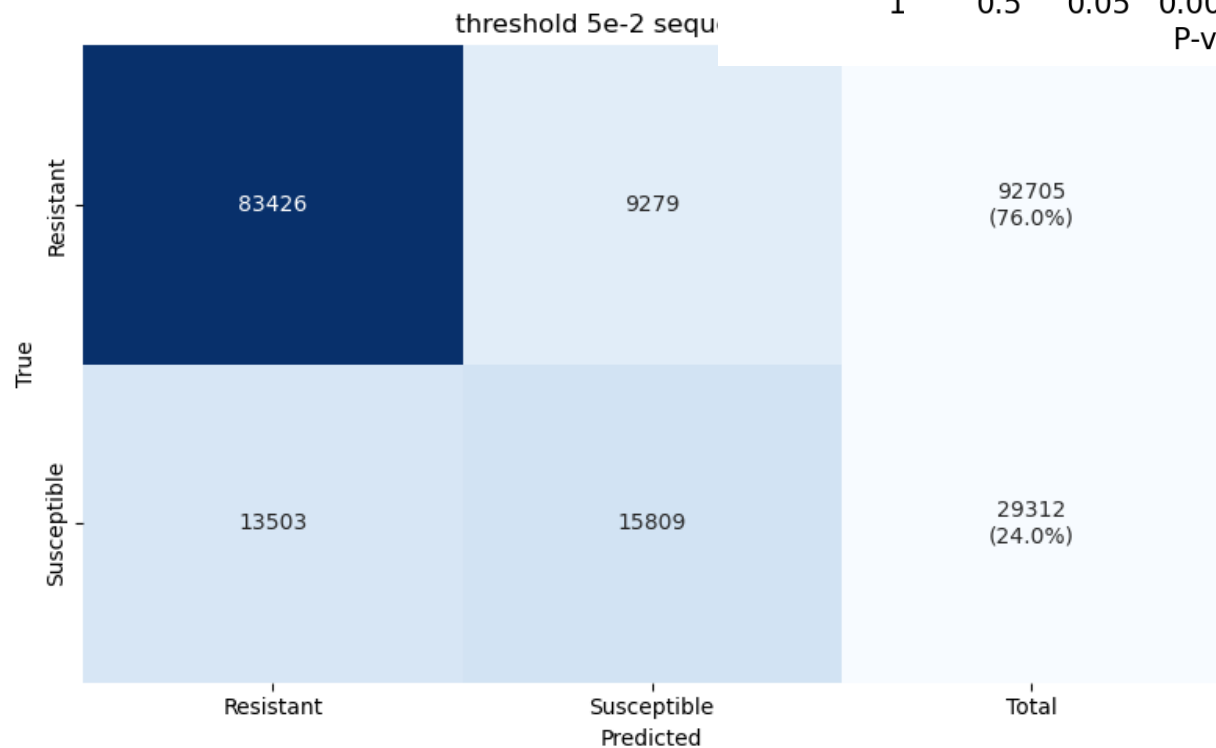
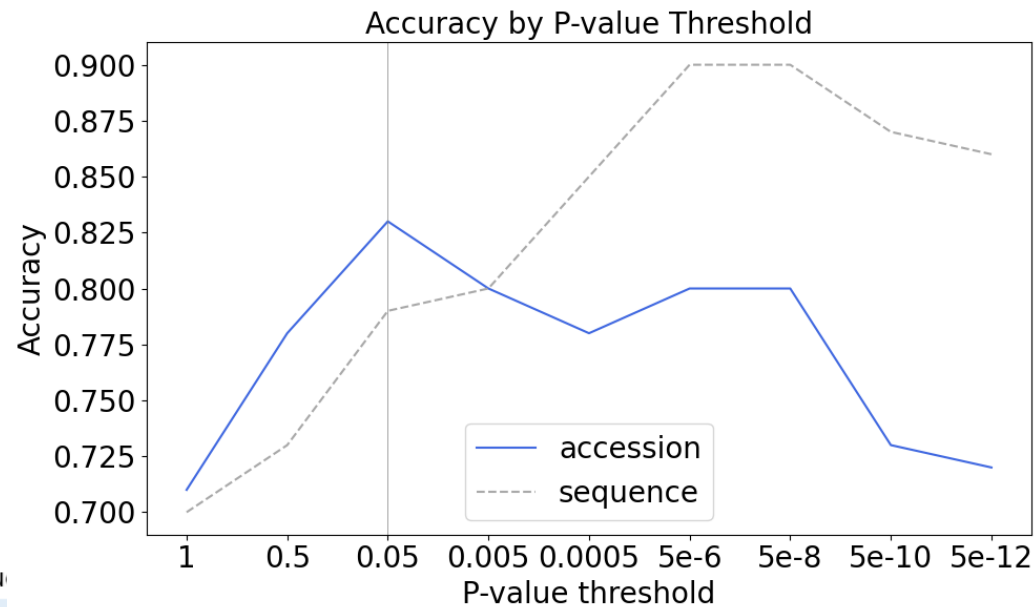
Biomedical Informatics MS student at Stony
Brook University



- As p-value decreases (more significant), the fraction of sequences that are **resistant** increases



- This makes the sequence-level dataset **imbalanced**, so accuracy is not a great metric here,
- Looking at the totals in the confusion matrix reveals this



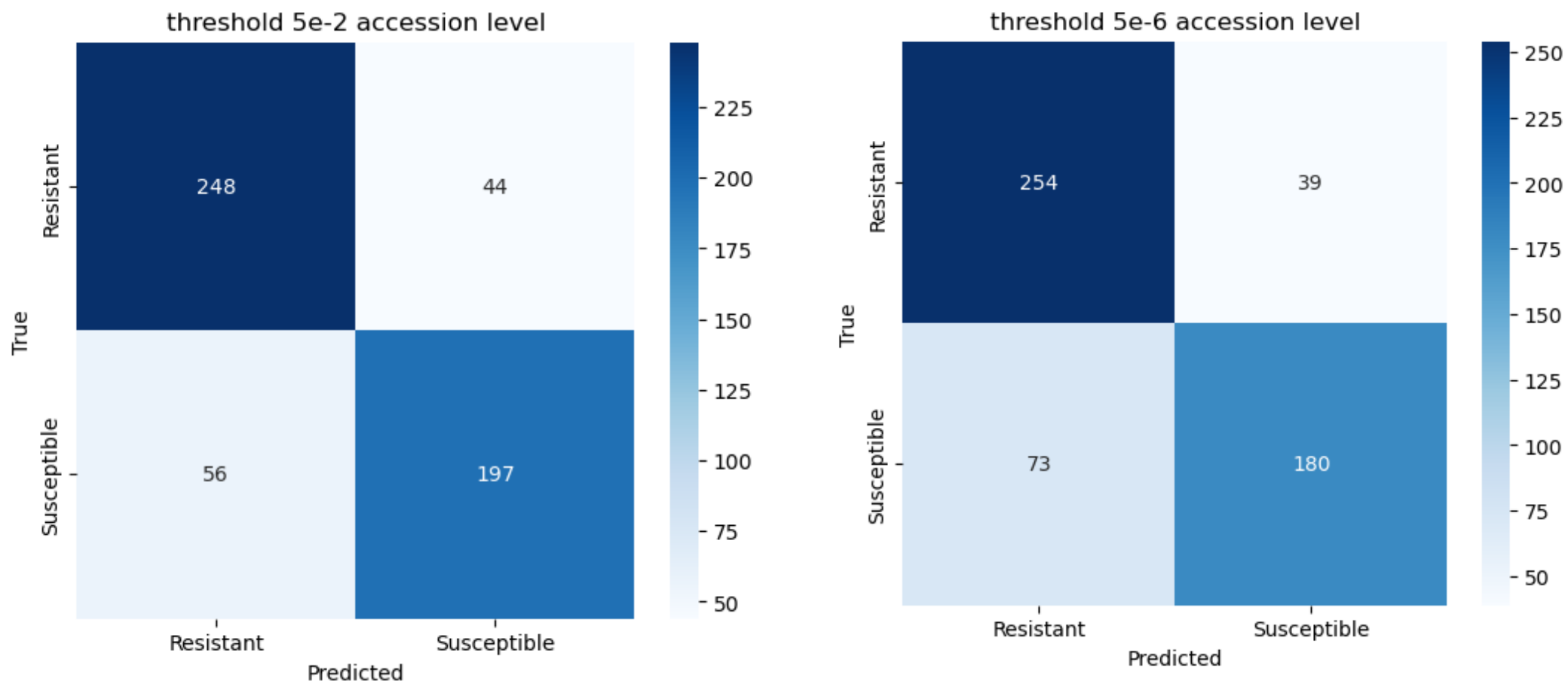
Weighted f1 still better:

.80

.90

36

- Accession level is **balanced** (accuracy is a **better metric** here than in sequence-level) – at a higher threshold resistant predicts a bit better but susceptible is much worse because there are **a lot less susceptible sequences** and a higher fraction of the remaining ones are **random** (from 0 num hits)



- My hypothesis is that at lower p-values, there is just less **voting power** for susceptible samples (more ties, less sequences to drown out outliers)